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Computers and Electronics in Agriculture

Date: 15 March 2026

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Deriving crop calendars from satellite phenology and unsupervised learning: an application to Andalusia (Spain)

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Highlights

- Sentinel-2 data used to derive phenology-based crop calendars in Andalusia.
- Remote sensing detects seasonal crop stages with 10m spatial resolution.
- Crop calendar validation using MAPA data across diverse agricultural types.
- Phenological clustering identifies six main seasonal crop groups.
- Circular statistics applied to phenology improves crop classification accuracy.

Abstract

Timely and accurate crop calendars are essential for optimizing agricultural practices and improving food security. Traditional calendar generation methods, based on field surveys, are often spatially limited and time-consuming. In this study, we present a replicable methodology to derive crop calendars using Land Surface Phenology (LSP) metrics extracted from Sentinel-2

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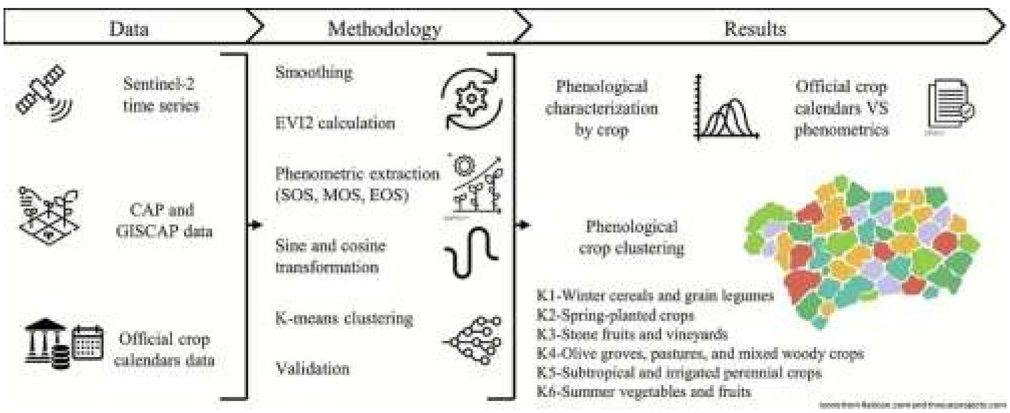
cropping systems. We computed Enhanced Vegetation Index 2 (EVI2) time series for more than one million agricultural plots and derived phenological metrics (start, middle, and end of season) using a double-logistic model. To address the circular nature of phenological data (expressed in Julian days), we applied sine and cosine transformations prior to clustering. Six phenological groups were identified using k-means, and a classification tree (CART) was used to interpret the clustering structure. The satellite-derived calendars were validated against official field-based calendars and showed high agreement for arable crops. The proposed approach allows for the generation of scalable, up-to-date, and spatially explicit crop calendars that can support agricultural monitoring systems, inform policy making, and enhance digital farming tools.

Graphical abstract



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Keywords

Crop phenology; Sentinel-2; Land Surface Phenology (LSP); Unsupervised classification; Crop calendar

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1. Introduction

The impact of climate change on agriculture is expected to reduce crop yields (Hidalgo, 2013).

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billion by 2100 (Lucas, 2012). In parallel, rising wealth levels are anticipated to drive increased per capita food consumption. Compounding these pressures, armed conflicts, political instability, and economic crises disrupt international agricultural trade and pose a significant threat to food security, especially in low-income countries (Food and Agriculture Organization of the United Nations, 2022, Zhou et al., 2023). In this context of growing uncertainty, agriculture must become both more efficient and more sustainable. Achieving this requires access to detailed, timely information to support planning and decision-making processes. One essential tool is the crop calendar, which represents the temporal sequence of phenological stages in a crop's growth cycle, spanning from the fallow period and land preparation through to crop establishment, maintenance, harvest, and storage (More et al., 2016, Yulianti et al., 2016). Crop calendars facilitate the optimization of land productivity through better-informed decisions, enhanced soil and nutrient conservation, and more effective pest and disease management. They are not only vital for farmers but also for market information systems and research on agricultural systems (Bondeau et al., 2007, Osborne et al., 2007, Stehfest et al., 2007). Moreover, crop calendar information serves as a key input for crop monitoring systems that provide early warnings of potential food security crises (Fritz et al., 2019). Prominent examples include: i) the Global Information and Early Warning System (GIEWS); ii) the Famine Early Warning Systems Network (FEWS NET); iii) the Monitoring Agriculture with Remote Sensing Crop Yield Forecasting System (MCYFS); iv) CropWatch; v) the USDA Foreign Agricultural Service (USDA-FAS); vi) the Group on Earth Observations Global Agricultural Monitoring Initiative (GEOGLAM); vii) the World Food Programme's Seasonal Monitor; and viii) the Anomaly Hotspots of Agricultural Production (ASAP). Among the various inputs used by these systems, crop calendar data is considered among the most critical (Fritz et al., 2019). It is also valuable for informing agricultural policy, although specific studies addressing this aspect remain limited. At the European level, the Common Agricultural Policy (CAP) represents the primary framework for supporting agriculture. It seeks to promote food security, improve farm productivity and rural livelihoods, preserve landscapes, and mitigate the effects of climate change on agriculture (European Commission, 2025). In this context, crop calendar data, alongside climate and crop stock information, can improve assessments of seasonal and spatial variability in food supply, and help quantify the risks posed by biotic and abiotic stressors during the crop cycle (Laborte et al., 2017). The heterogeneity of agricultural systems across the European Union, together with inconsistencies in national crop classification systems, poses a challenge for harmonizing crop types into standard categories. This is particularly problematic in areas with mixed cropping systems or loosely defined crop groupings. Some research initiatives aim to develop objective classification schemes that go beyond botanical criteria to include agricultural practices and end products, thereby enabling greater standardization for CAP implementation (European

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classification, aligning crops by phenological stages rather than strict botanical traits. Grouping by calendars would enhance consistency and comparability in CAP implementation.

Historically, crop calendars have been developed using information collected by farmers or agricultural experts through field campaigns. However, these observations tend to focus on specific crops or regions, are often aggregated at the national level, or are subject to inconsistencies due to differing methodologies and observer interpretations (Sacks et al., 2010). Moreover, such efforts are time-consuming, costly, and vulnerable to discontinuity. In contrast, remote sensing offers a promising alternative. Most crops exhibit distinct phenological activity at different stages of the growing season, which can be captured via satellite data. The study of such seasonal dynamics is known as phenology (Lieth, 1974), and when derived from satellite observations, it is referred to as Land Surface Phenology (LSP). LSP enables the characterization of vegetation cycles and is especially useful for developing crop calendars, thereby overcoming the limitations of traditional field-based methods. LSP is typically derived from time series of satellite-based vegetation indices (VIs) or biophysical variables, which act as proxies for plant vigour and biomass (Caparros-Santiago, 2021).

Developing crop phenology monitoring systems at regional scales requires satellite data with wide spatial coverage, high revisit frequency, high spatial resolution suitable for typical plot sizes, as well as affordable data acquisition (Carreño-Conde et al., 2021, Graf et al., 2022, You et al., 2013). In this context, PlanetScope imagery stands out as a promising option due to its high spatial resolution and revisit frequency, although its high cost remains a significant limitation for large-scale or long-term monitoring (Skoures et al., 2021). The Copernicus Land Monitoring Service's High Resolution Vegetation Phenology and Productivity (HR-VPP) dataset provides high-spatial (10 m) and high-temporal resolution vegetation phenology and productivity metrics across Europe, offering a valuable benchmark for evaluating and complementing Sentinel-2 based phenological analyses (Tian et al., 2021). HR-VPP is primarily designed for natural vegetation, relying on Gross Primary Productivity (GPP) based modelling calibrated with carbon flux tower measurements. Consequently, it is not specifically optimised for detecting crop phenophases The MODIS sensor offers daily global coverage but is limited by its coarse spatial resolution (≥250m), which restricts its use for plot-level analyses (Wardlow et al., 2007, Inglada et al., 2015). By contrast, the Sentinel-2 mission provides high-resolution imagery (up to 10m) every five days, making it highly suitable for phenological studies that require detailed temporal monitoring. Cloud-free observations are particularly important during such phases (Banskota et al., 2014, Inglada et al., 2015). Full-season coverage, including pre and post growing periods, is essential to reliably capture crop dynamics. The maturation phase is particularly valuable for

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help differentiate crop types and identify phenological anomalies by comparing individual seasons to a reference year, potentially indicating yield losses. Time series of VIs allow for the extraction of phenological metrics (hereafter, phenometrics) such as the start of season (SOS), middle of season (MOS), end of season (EOS), and length of season (LOS), which form the basis of satellite-derived crop calendars (Caparros-Santiago, 2021). The normalized difference vegetation index (NDVI) is the most commonly used VI, though it tends to saturate in areas with dense vegetation. The enhanced vegetation index 2 (EVI2) offers improved sensitivity under such conditions, while also reducing atmospheric and soil background effects (Claverie et al., 2018). Studies have shown that EVI2 is more responsive to vegetation type variation and can improve yield estimations (Garcia-Perez et al., 2023, Iglesias et al., 2012, Büttner et al., 2004). Several methods exist to derive phenometrics from VI time series (Shen et al., 2023). These include: (i) threshold-based methods, which identify phenological stages based on fixed VI values or relative percentages of the seasonal amplitude (Verger et al., 2016, Sisheber et al., 2022, White et al., 2009, Bórnez et al., 2020); (ii) shape-model fitting approaches, which integrate ground and satellite observations using scalable phenology models (Liu et al., 2022, Sakamoto, 2018, Sakamoto et al., 2010); (iii) curvature-based models, which detect inflection points in the VI curve without relying on predefined thresholds (Zhang et al., 2003, Hermance et al., 2007, Zhou et al., 2015) and (iv) moving average method, in which the VI value is compared to its moving average to determine changes in vegetation trends (Reed et al., 1994). The use of circular statistics and specifically sine and cosine transformations of phenological stages remains relatively uncommon in large-scale agricultural phenology studies. While some ecological research has demonstrated the value of these transformations to preserve the cyclicity of variables (Fisher et al., 2007, Morellato et al., 2010), their application in crop phenology is still limited.

Crop calendars can be developed by governmental institutions such as the Spanish Ministry of Agriculture, Fisheries and Food (MAPA, Spanish acronym) (Ministerio de Agricultura, Pesca y Alimentacion, 2021) or the United States Department of Agriculture (USDA) (USDA), 2010, USDA, 2025a, USDA 2025b). They can also be developed by international organizations, such as the Food and Agriculture Organization for the United Nations (FAO) (Food and Agriculture Organization for the United Nations, 2025a), the Joint Research Centre of the European Commission (Dimou, Meroni, & Rembold, 2018), or the International Fertilizers Association (International Fertilizer Association, 2025). These calendars vary in terms of the level of detail regarding the crops and varieties considered, the spatial aggregation scale, and the countries or regions covered. The scientific literature identifies the field calendars developed by USDA-FAS (United States Department of Agriculture, 2025a) and FAO (Food and Agriculture Organization of

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or regional level, which limits its utility for capturing local agricultural specificities. The FAO calendar offers more detailed information, distinguishing ecoregions within different countries, but it is only available for selected developing nations. This calendar is based on crop analysis information and provides data on planting and harvesting dates for 130 crops, with a temporal resolution of 15 days (Franch et al., 2022). There is a gap between large-scale studies at the continental or national level and localized studies that focus on small areas with low crop diversity and relatively homogeneous land conditions (Franch et al., 2022, Whitcraft et al., 2015, Descals et al., 2021, Helman, 2018, Bórnez et al., 2020, Zhao et al., 2024). Bridging this gap requires the development of regional-scale crop calendars, particularly at the NUTS-2 level in Europe, where agricultural landscapes exhibit significant heterogeneity. Andalusia, the largest region in Spain and one of the most important agricultural areas in Europe, has been the subject of numerous phenological studies, particularly focused on the olive tree due to its economic significance. However, there is no comprehensive phenological framework encompassing the full agricultural diversity of the region (Garcia-Mozo et al., 2010, Medina-Alonso et al., 2020, Oteros et al., 2013).

This study pursues two main objectives: first, to develop a reference methodology for systematically generating crop calendars from remote sensing-based phenology data; and second, to conduct a detailed phenological characterization of crops in Andalusia—a European NUTS-2 region characterized by high agroclimatic diversity. The proposed methodology also aims to group crops based on their LSP behaviour, beyond conventional botanical classifications.

2. Study area, data and methodology

2.1. Study area

Andalusia, the southernmost region of Spain and one of the largest in Europe, covers a total area of 87,599 km^2. In terms of land area, Andalusia would rank as the 19th largest country in Europe and the 14th within the EU, comparable in size to Portugal, Azerbaijan, or Austria. The region's geographical position at the southwestern edge of Europe makes it strongly influenced by the Atlantic Ocean, the Mediterranean Sea, and the Sahara Desert (Fig. 1). The Mediterranean basin has been identified as one of the most vulnerable areas to climate change (IPCC, 2023), which directly impacts Andalusia’s agricultural systems. The physical environment of Andalusia is highly heterogeneous, structured into three main geomorphological units: (i) Sierra Morena, a moderately elevated, eroded mountain range in the north; (ii) the Guadalquivir Valley, a fertile depression in the central region; and (iii) the Betic Systems, a series of mountain ranges in the

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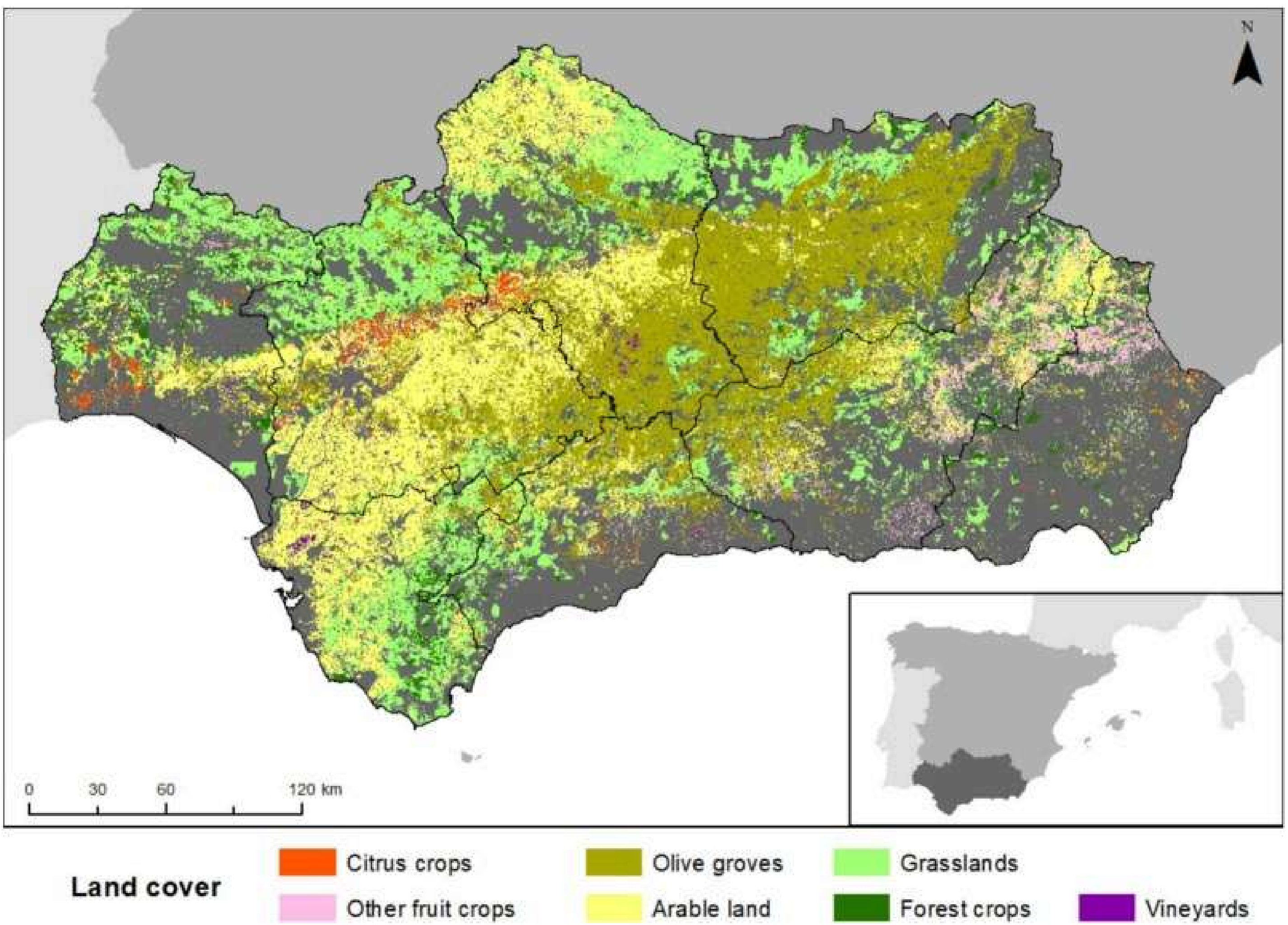
has poor soils unsuitable for most crops, with agricultural activity mainly limited to pastureland. In contrast, the Guadalquivir Valley supports the highest diversity of crops due to its fertile soils and access to irrigation. Irrigated lands near the Guadalquivir and Genil rivers are dominated by rice fields and orange groves, while rainfed areas are used for cereals and industrial crops such as sunflower and cotton. The surrounding countryside hosts extensive olive groves, positioning Andalusia as the world's leading olive oil-producing region. The Betic Systems, with their steep terrain and high altitudes, create localized climatic conditions where agricultural activity is more limited, though some valleys support specialized crops. Along the coast, although agriculture covers a smaller area, it plays a crucial socio-economic role. In the western coastal zone, strawberry and other red fruit production under plastic greenhouses is prominent. In the central Mediterranean area, subtropical crops such as avocado, mango, and custard apples thrive due to the unique climatic conditions. The eastern coast, characterized by low annual precipitation, hosts extensive greenhouse farming for vegetables, fruits, and flowers.



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The study area encompasses 34,649 km2, representing 39.55% of Andalusia's total land area, which is the percentage subsidized through the CAP. It includes most of the region's agricultural land, covering forestry, pasture, and fallow land. The agricultural landscape is dominated by olive groves, which cover 1,142,427.98 ha across 739,897 plots, resulting in a highly fragmented structure with an average plot size of 1.54 ha, according to CAP-GISCAP data (Table 1). Almond groves extend over 142,677.65 ha, distributed across 88,463 plots, with a mean plot size of 1.61 ha. Among arable crops, sunflower is the most widespread (103,551.40 ha), followed by durum wheat (94,589.34 ha) and barley (84,343.22 ha), with average plot sizes ranging from 2.51 to 4.04 ha. Other cereals, such as common wheat (76,511.86 ha), oat (53,961.46 ha), and triticale (30,258.38 ha), also occupy significant areas. Chickpea fields cover 14,454.19 ha, with an average plot size of 3.65 ha. Rice plots span 34,477.51 ha, with a mean plot size of 8.21 ha, suggesting a more consolidated distribution. Cotton covers 31,313.77 ha, with smaller, more fragmented plots averaging 3.78 ha. Orange groves extend over 38,054.50 ha, with an average plot size of 2.61 ha.

Table 1. Crops included in the main body of the study according to CAP-GISCAP data.

Crop	Number of plots	Total area (ha)	Plot mean area (ha)	Plot area standard deviation (ha)
Olive grove	739,897	1,142,427.98	1.54	4.43
Almond grove	88,463	142,677.65	1.61	3.37
Sunflower	27,838	103,551.40	3.72	8.03
Durum wheat	23,440	94,589.34	4.04	8.88
Barley	33,546	84,343.22	2.51	5.06
Common wheat	23,651	76,511.86	3.24	6.8
Oat	26,735	53,961.46	2.02	4.54
Orange grove	14,601	38,054.50	2.61	5.24
Rice	4,201	34,477.51	8.21	6.51
Cotton	8,292	31,313.77	3.78	5.38
Triticale	9,636	30,258.38	3.14	6.34
Chickpea	3,959	14,454.19	3.65	7.27

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2.2. Data and methodology

Several data sources were used in this study: (i) Sentinel-2 satellite images (Level-2A) for the 2019 agricultural season, (ii) CAP farmer declarations, (iii) GISCAP plot locations, and (iv) MAPA crop calendars for sowing and harvesting dates. The methodology consisted of the following steps: (i) integration of CAP and GISCAP datasets and selection of plots, (ii) acquisition of Sentinel-2 images from July 1, 2018, to June 30, 2020, with a revisit period of five days, (iii) smoothing of the time series and gap filling using a double-logistic function, (iv) calculation of EVI2, (v) extraction of the mean EVI2 value for each plot, (vi) derivation of plot-level phenometrics, (vii) validation of results against field calendars, and (viii) clustering of crops based on phenometrics using the k-means algorithm (MacQueen, 1967).

All agricultural plots in Andalusia were initially considered with the exception of crops covering less than 400 ha. From the remaining dataset, the 12 most widespread crops by area—identified both through CAP declarations and the MAPA crop calendar—were included in the main body of the study. Major CAP categories such as permanent grassland or fallow land were omitted, as they do not constitute actual crops.

Sentinel-2 imagery from the Copernicus programme was used for temporal analysis and was retrieved from ESA's Sentinel Scientific Data Hub using the 'sen2r' R package (Ranghetti et al., 2020). The processed Level-2A bottom-of-atmosphere (BOA) reflectance images spanned from 1 July 2018 to 30 June 2020, thus enabling the full crop growth cycle to be captured, even if it did not fall entirely within the central year. The choice of the 2019 season was based on its inclusion in the most recent MAPA crop calendar, valid for five years, and on the comprehensive availability of Sentinel-2 imagery for that year. In contrast, the 2014–2016 period used to compile the MAPA crop calendar lacked full Sentinel-2 constellation operation, resulting in reduced image frequency. The Sentinel-2 bands used to compute the EVI2 time series were Band 4 (red) and Band 8 (near-infrared) (European Space Agency, 2015), both with a spatial resolution of 10 m. Each tile had two available orbits, with priority given to the one covering the largest area; in cases where neither orbit provided full coverage, both were incorporated (Fig. 2). To minimize gaps in temporal information for plots located in areas of orbit or tile overlap, the orbit with the highest number of available observation dates was prioritized. EVI2 values were extracted for all available images and smoothed using the double logistic model proposed by Zhang et al. (2003). Nevertheless, the logistic function was used only as a smoothing technique. Bórnez et al. (2020) found that the threshold method was the most effective for phenometric extraction. The start of season (SOS) was identified as the day of the year (DOY) when EVI2 exceeded a predefined relative threshold equal to 15% of the seasonal amplitude, while the end

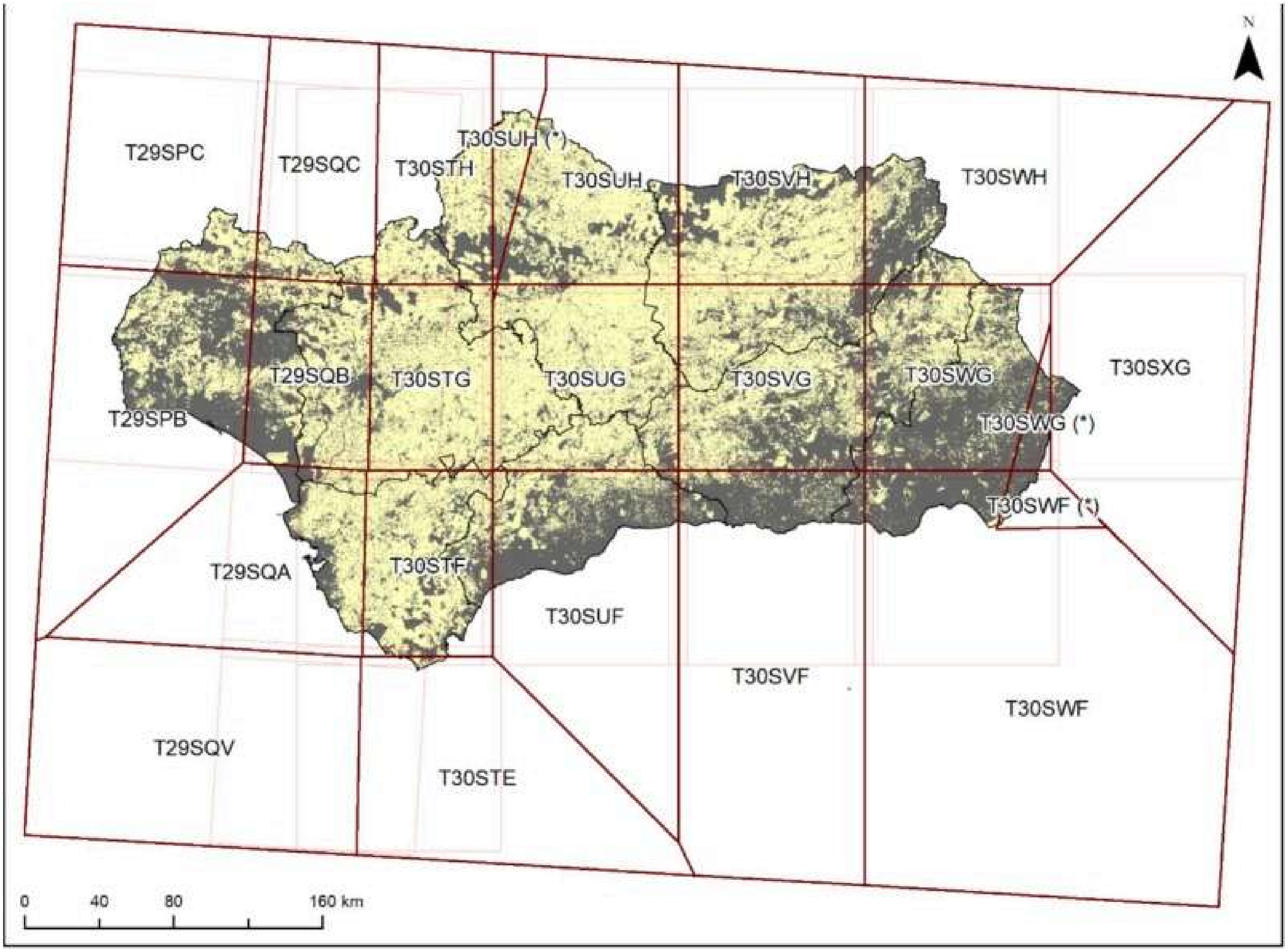

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Fig. 2. Sentinel-2 tiles included. Tiles marked with an asterisk (*) used the secondary orbit data.

CAP declarations provided comprehensive information on each plot, including crop type, variety, irrigation system, plot area, and the portion of the plot under cultivation. For practically all plots, information was provided for only a single crop per year. These records were linked to the Spanish SIGPAC system, a geospatial database defining the geometry and location of each plot. This linkage enabled the extraction of pixel-level EVI2 values for each plot, facilitating the calculation of plot-specific and crop-specific averages. The MAPA crop calendar (MAPA, 2021) contained phenological data for 159 crops, categorized into twelve families according to the CNAE-2009 classification. These included seven herbaceous families (cereals, legumes, tubers, industrial crops, forage crops, vegetables, and ornamental plants) and five woody crop families (citrus, non-citrus fruit crops, vineyards, olive groves, and other woody crops). Published in

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and regional (NUTS-2) levels. For this study, NUTS-3 level data were used to maximize spatial resolution and ensure the highest level of detail. The MAPA crop calendar provided monthly percentages of sown or flowering areas (depending on whether the crop was herbaceous or woody), as well as harvested and marketed production. It also reported total cultivated area and cumulative production, following Regulation (EC) 543/2009 (European Parliament and Council of the European Union, 2009). The MAPA crop calendar served as the main benchmark for validating the crop phenological trajectories and phenometrics derived from LSP time series. This comparison ensured alignment between satellite-based estimates and officially reported agricultural dynamics in Andalusia.

Agro-phenological zones were defined using the k-means clustering algorithm, a partition-based method (Rivera et al., 2022). The variables included in clustering were SOS, MOS (middle of season), and EOS. Given the broad diversity of crops, clustering was conducted separately: (i) arable crops, (ii) woody crops, and (iii) also for all crops, incorporating the herbaceous/woody classification as an additional feature. To quantify the differences in phenological behaviour among crops, we conducted a one-way analysis of variance (ANOVA) for each phenological metric (SOS, MOS and EOS), considering the crop type as the grouping factor. Prior to the ANOVA, phenological dates were expressed as day of year (DOY). To visualise the distribution of values for each crop, violin plots with points were generated.

It is important to note that satellite-observed phenology (LSP) may diverge substantially from ground-based observations, highlighting the need to define meaningful LSP-based groupings that can fully exploit the capabilities of current and future Earth observation programs for agricultural monitoring. A hybrid, semi-supervised approach is proposed for crop calendar development, that integrates clustering and supervised classification techniques to address the challenges inherent to phenological data. To group crops into phenologically coherent families, an unsupervised k-means clustering algorithm was applied. However, since phenometrics were expressed in Julian days, their circular nature renders standard Euclidean distances inappropriate, as they fail to account for the continuity between the end and beginning of the calendar year. To overcome this limitation, temporal variables were transformed using their sine and cosine components, thereby preserving their cyclical structure and enabling the accurate clustering of phenological stages, following recommendations from Morellato et al. (2010), yielding six variables for clustering. The elbow method was employed to determine the optimal number of clusters by identifying the point beyond which additional clusters provided diminishing differentiation, as cluster membership became less distinct (Rodriguez-Galiano et al., 2022). To avoid overrepresentation of more dominant crops, a stratified random sampling required number. A classification and regression tree (CART) was subsequently employed to derive interpretable rules for each cluster. This step not only resolved the circularity issue but also ensured that phenological groups respected the temporal continuity of the original metrics. The CART was then trained on the k-means results to derive phenological classification rules for each crop category, offering insight into representative seasonal dynamics in the region.

3. Results and discussion

3.1. Regional phenological dynamics of crops in Andalusia

The methodology developed enabled an exhaustive characterization of crop phenology in Andalusia, providing specific dates for SOS, MOS, and EOS, as reported in other regions through studies with related research objectives (Li et al., 2020; Bórnez et al., 2020, Franch et al., 2022). Phenometrics were derived at the plot level across the entire region (Fig. 3, Fig. 4, Fig. 5). The Mediterranean climate prevailing in Andalusia largely explains the broadly homogeneous phenological behavior observed across much of the territory, with SOS generally occurring during the final months of the year, coinciding with the return of rainfall after the summer drought and the onset of cooler temperatures. Most crops reached their peak vegetative activity during the winter and early spring, completing their cycles in late spring or, more predominantly, during the summer months. More than 65% of the plots exhibited SOS during the last quarter of the year, with 35% initiating in October. Nearly 70% of the plots reached MOS during the first quarter, with February accounting for 26% of cases. Regarding EOS, almost 50% of plots completed their cycle in the second quarter, while 40% did so in the third quarter, with June being the most representative month at nearly 34%. These results reflect a certain degree of phenological homogeneity across the region, attributable to its Mediterranean climate, where the initiation of vegetative activity typically coincides with the return of autumn rainfall and the onset of milder temperatures following the summer drought. Nevertheless, this pattern was not entirely uniform, with secondary peaks associated with summer crops dependent on irrigation systems. A representative example of this variability is observed in the agricultural zones surrounding the Guadalquivir River, particularly the rice fields in the Lower Guadalquivir, which exhibited SOS around June, MOS in August, and EOS in October (Fig. 3, Fig. 4, Fig. 5).

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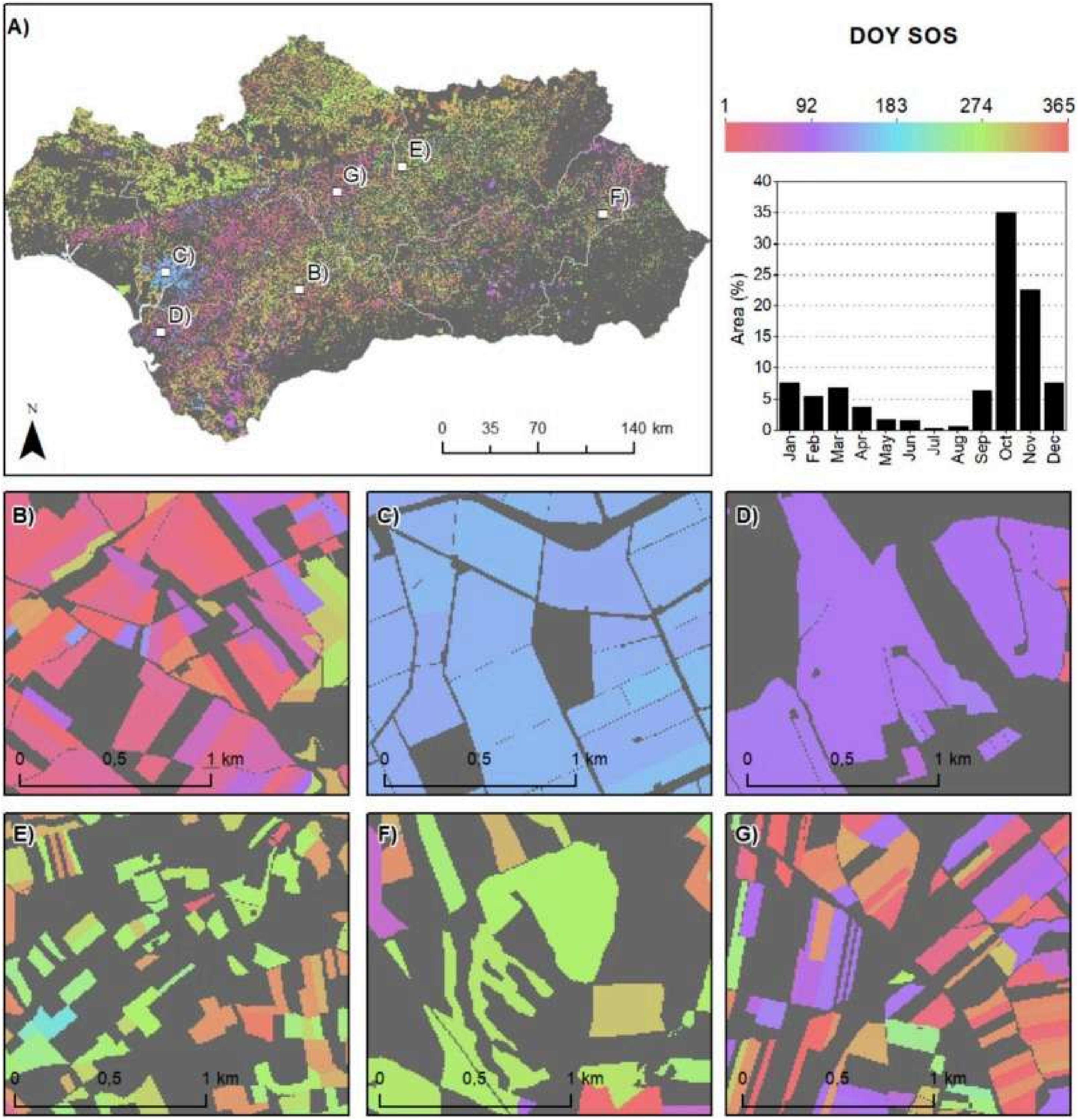

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Fig. 3. DOY in which SOS was reached in each plot. A) Andalusia; B) Cereal plots; C) Rice plots; D) Vineyards; E) Olive groves; F) Almond groves; G) Mixture of sunflower plots and other non-irrigated crops. The graph represents the percentage of area which reached SOS in each month.

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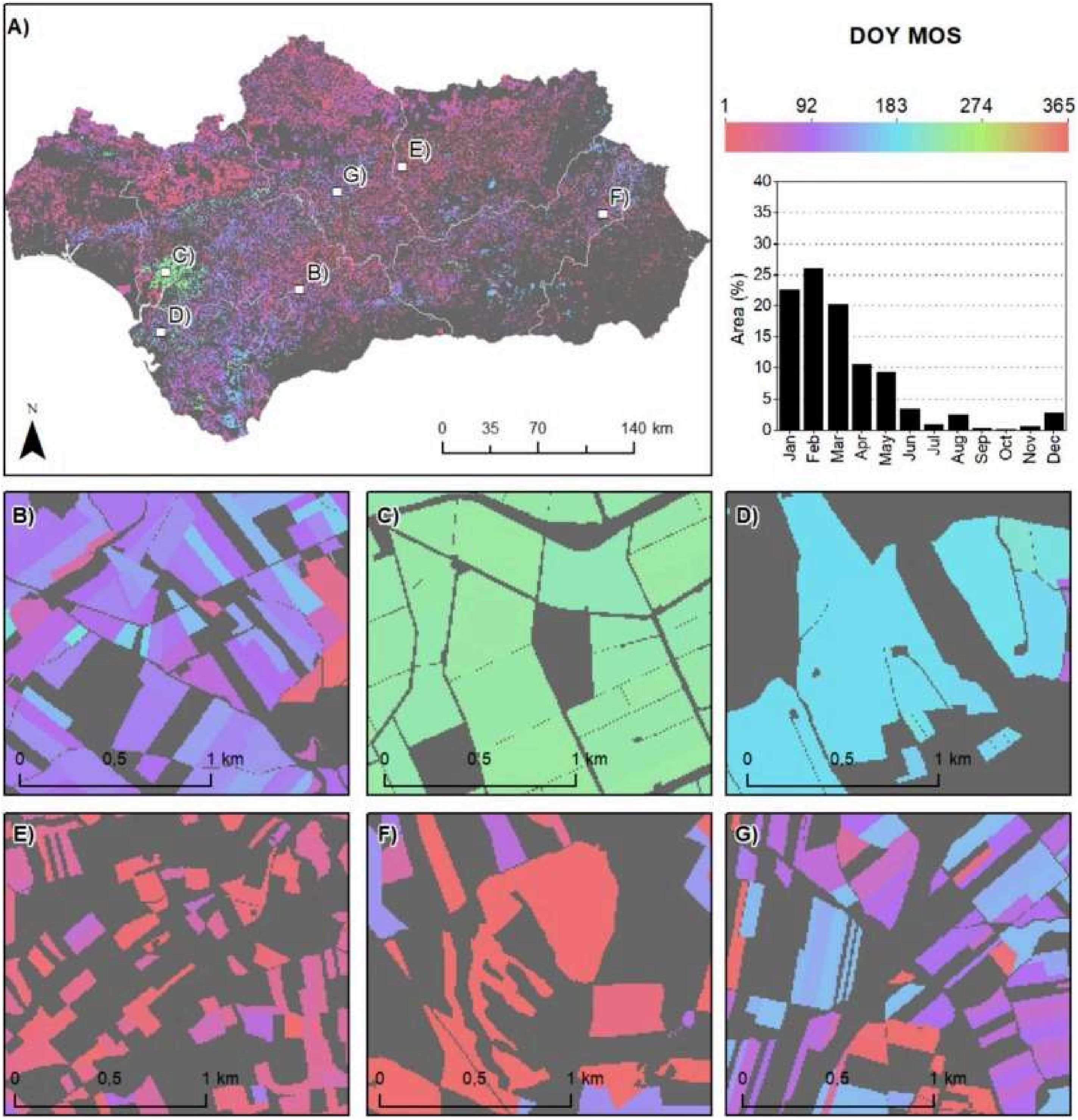

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Fig. 4. DOY in which MOS was reached in each plot. A) Andalusia; B) Cereal plots; C) Rice plots; D) Vineyards; E) Olive groves; F) Almond groves; G) Mixture of sunflower plots and other non-irrigated crops. The graph represents the percentage of area which reached MOS in each month.

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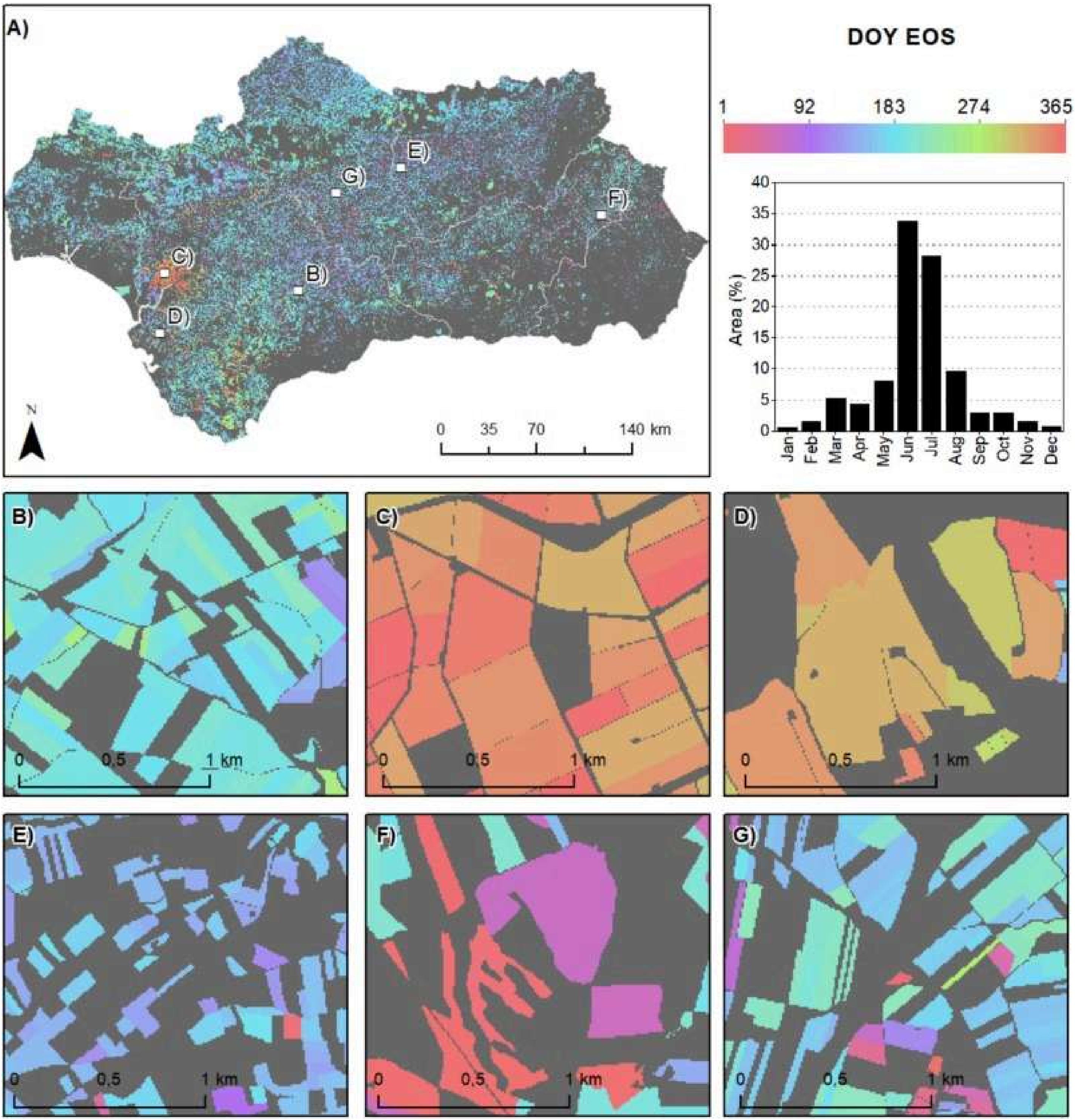

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Fig. 5. DOY in which EOS was reached in each plot. A) Andalusia; B) Cereal plots; C) Rice plots; D) Vineyards; E) Olive groves; F) Almond groves; G) Mixture of sunflower plots and other non-irrigated crops. The graph represents the percentage of area which reached EOS in each month.

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The detailed results at the crop level revealed a considerable disparity in phenological behavior. Differences were observed not only between different crop types, but also, in many cases, within the same crop, reflecting intra-specific variability influenced by local environmental and management conditions, as previously documented in other regions (Mugabowindekwe et al., 2018). The temporal trajectories derived in our study might help to highlight crop-specific phenological events, like Duan et al. (2024), who extracted phenological events of apple orchards. Due to the large number of crops included in the study, detailed information for each crop is provided in the supplementary material, while only the most spatially extensive crops are presented here (Fig. 6). High values of EVI2 indicate periods of maximum vegetative activity, whereas low values correspond to recently sown crops or even the absence of vegetation. The analysis of off-season behaviour showed high standard deviation values, primarily due to plot-level crop rotations—except in the case of typically permanent crops, particularly woody species such as olive groves or cork oak woodlands. Arable crops exhibited well-defined EVI2 peaks corresponding to periods of maximum phenological activity and, therefore, the highest VI values. For example, durum wheat and barley reached maximum EVI2 values close to 0.8 in April and May, coinciding with their peak vegetative growth before harvest. Similarly, crops such as common wheat, cotton, and sunflower exhibited distinct peaks between May and July, with EVI2 values ranging from 0.6 to 0.9. Phenological similarities among crops belonging to the same botanical family have been extensively documented in the scientific literature (Ghazaryan et al., 2018, Salinero-Delgado et al., 2022). In contrast, woody crops such as olive, almond, and orange displayed smoother phenological trajectories. Olive groves, for instance, maintained EVI2 values between 0.2 and 0.4 throughout most of the year, with slight increases in late autumn and early winter, reflecting lower seasonal variability compared to arable crops. Almond trees showed moderate variation, with EVI2 values between 0.3 and 0.5, indicating a more continuous vegetative cycle with a mild increase in spring that aligns with flowering. The homogeneity of EVI2 values, as represented by standard deviation, was also a relevant aspect. In general, arable crops exhibited greater variability across the time series, with higher standard deviations during periods of intense phenological activity and at the temporal extremes of the cropping season. This pattern reflects the inclusion of both the preceding and following cropping seasons, in which different crops may have been cultivated due to rotation. In contrast, woody crops exhibited a more stable standard deviation profile across the year, consistent with their more uniform phenological trajectories. The characterization of these trajectories could support crop type discrimination throughout the season, as reported in other studies (Wang et al., 2024).

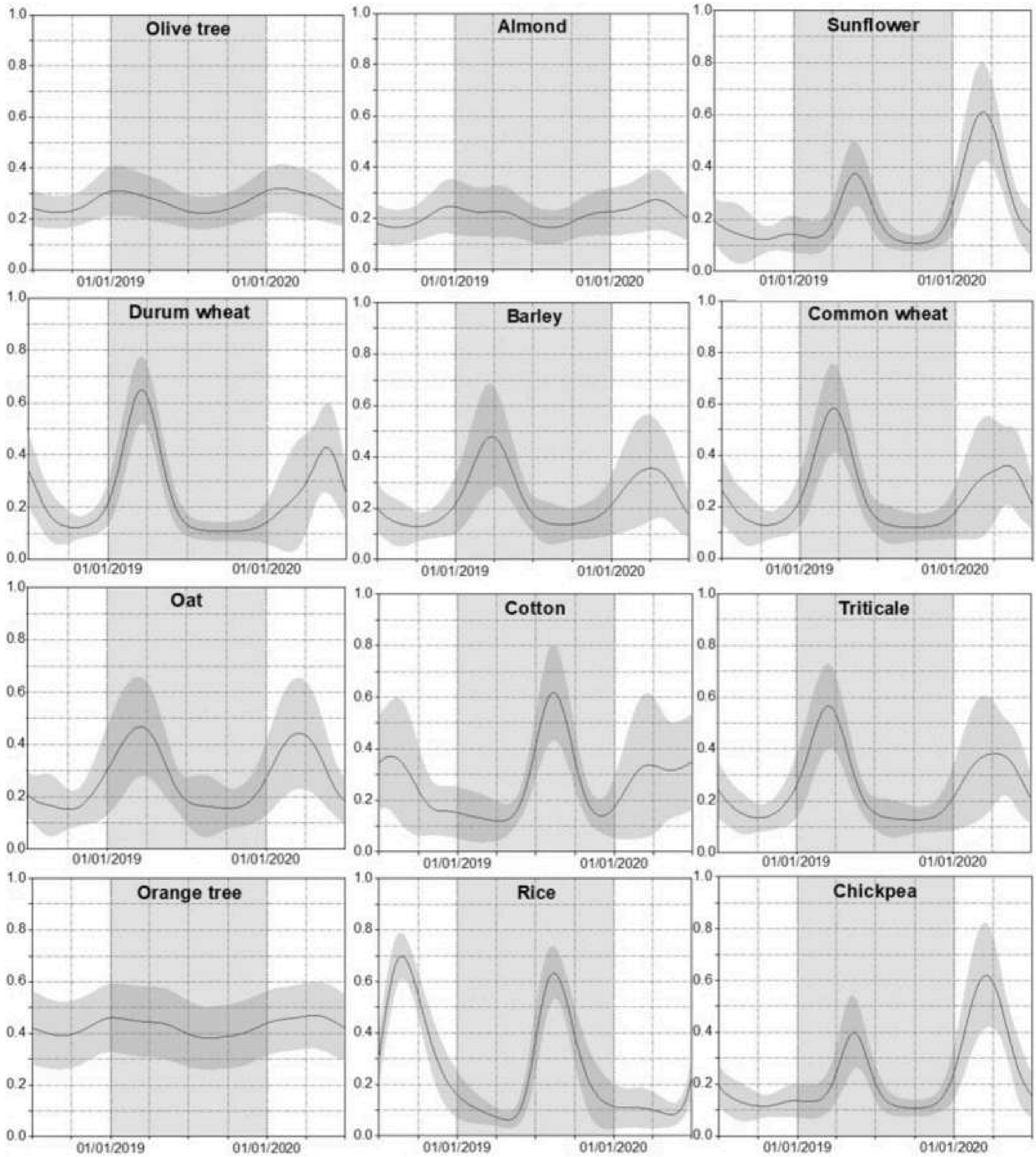

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Fig. 6. Seasonal phenological trajectories of EVI2 between 1 July 2018 and 30 June 2020. The

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The analysis of variance (ANOVA) revealed highly significant differences among crops for the three phenometrics (SOS, MOS and EOS) ($p<0.001$ in all cases) (Table 2). These differences reflect the distinct temporal dynamics characteristic of each crop type. Violin plots showed that the distributions of SOS, MOS and EOS are strongly concentrated for most crops, although species whose phenometrics occur around the transition between years (e.g., SOS in winter cereals) exhibit artificially widened distributions due to the circularity of day-of-year (DOY) values. This behaviour inflates the apparent variability of some crops despite their actual phenological similitudes. Traditional boxplots could not account the circularity of phenometrics, thus, violin plots were used instead (Fig. 7). Overall, both the ANOVA and the distributional plots corroborate that the three metrics provide robust information for separating the phenological behaviour of the studied crops.

Table 2. One-way analysis of variance (ANOVA) results for phenological metrics across crop categories.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
SOS	11	1,124,319,832	102,210,894	11622.39	0
Residuals SOS	361,926	3,182,888,960	8794.31	–	–
MOS	11	411,870,181	37442743.8	5531.58	0
Residuals MOS	361,926	2,449,841,414	6768.9	–	–
EOS	11	300,525,242	27320476.5	14560.72	0
Residuals EOS	361,926	679086871.7	1876.31	–	–

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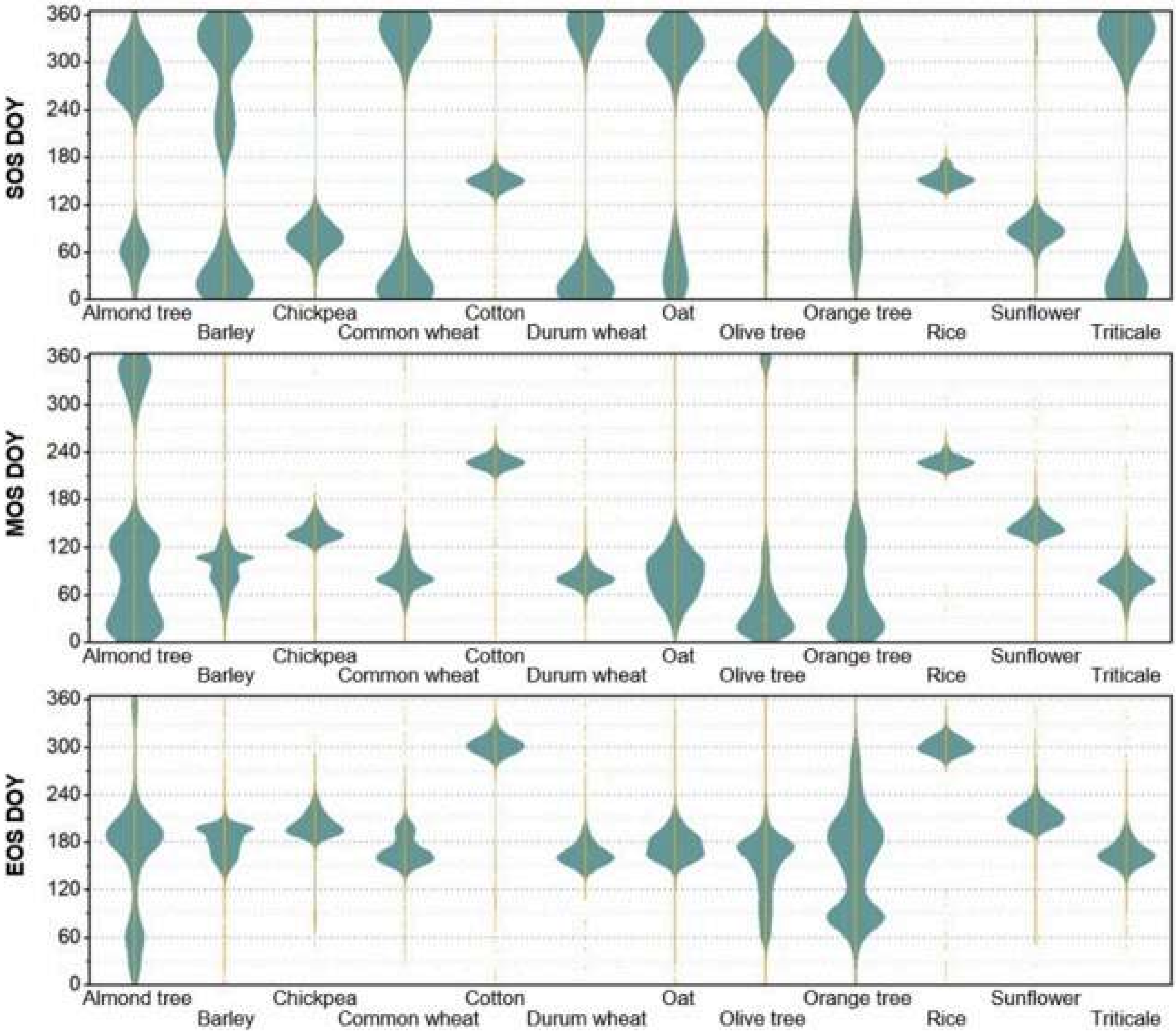

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Fig. 7. Distributional violin plots of crop phenology metrics.

3.2. Comparison between satellite-derived phenology and MAPA crop calendars

The results obtained from LSP were compared with MAPA crop calendar data. This allowed for the validation of the results, improving methodologies similar to the one proposed by Boori et al. (2019). Other studies included validation using field phenology data (Mascolo, Martinez-Marin, & Lopez-Sanchez, 2025) but focused on a specific area and crop. The use of crop calendar data provided information for all crops present in Andalusia. Due to the large agricultural diversity, it was difficult to establish general conclusions for all crops. As demonstrated by Mugabowindekwe et al. (2018), significant variability in maize crop calendars was observed

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environmental conditions on planting and harvesting dates. Given Andalusia's broader climatic and geographic diversity, an even greater heterogeneity in crop calendar dynamics is to be expected within the region. However, some common patterns were observed (Fig. 8), especially for arable crops. For these latter crops, a general coincidence was found between SOS and sowing, as well as between EOS and harvest. Winter cereals such as durum wheat, common wheat, barley, oats and triticale showed similar phenological trajectories. The sowing date for winter cereals was mainly concentrated between November and December, with November being the main month for triticale, oats and barley and December for durum and common wheat. The date at which SOS was reached was concentrated in three months, but in this case from November to January. November was the month with the most sowing for oats, December for triticale, and January for durum wheat, common wheat and barley. For wheat data it represented a temporal delay compared to what was stated by Franch et al. (2022), which could be explained by the global scale of that study. Overall, it can be said that there was a time gap of one month between the sowing date and the SOS. This was due to the natural delay that occurred between sowing and the moment when the crop germinated, allowing SOS to be identified through remote sensing, as noted in other studies such as Zhang et al. (2017). Regarding the harvest date, the behaviour of winter cereals was also very similar. In summer annual crops, sowing was reached between March and April for chickpea and sunflower, and in May-June for rice and cotton, while SOS dates were around March for chickpea, March-April for sunflower, cotton in May-June and rice in June. Harvesting was also heterogeneous, with chickpea harvested mainly in June-July, sunflower in July-August, and rice and cotton in October-November. These dates were similar in all four crops to those of EOS. In general, and despite these differences, motivated by the characteristics of each crop, the arable crops generally showed a high agreement between EOS and harvest, indicating that the end of vegetative activity coincided directly with grain harvest, a common feature in annual crops. Moreover, SOS and sowing also tended to coincide or be very close in time, which highlighted the synchronisation between the start of the phenological cycle and the agronomic action of sowing in these crops. On the other hand, woody crops showed notable differences with respect to annual crops. As they did not have to be planted every season, they presented a smoother and more progressive phenological trajectory compared to herbaceous crops in general (Fig. 8), although of longer duration, with almond and orange trees reaching the SOS around October and ending the season in June-July. Olive trees had a shorter growing season, starting in December-January and ending in June. The calendars provided information on flowering and not on sowing due to the logical nature of woody crops, which resulted in a less obvious correspondence between phenology and calendar data. The only exception to this fact was an apparent simultaneity between harvest and SOS, which was particularly clear for olive and

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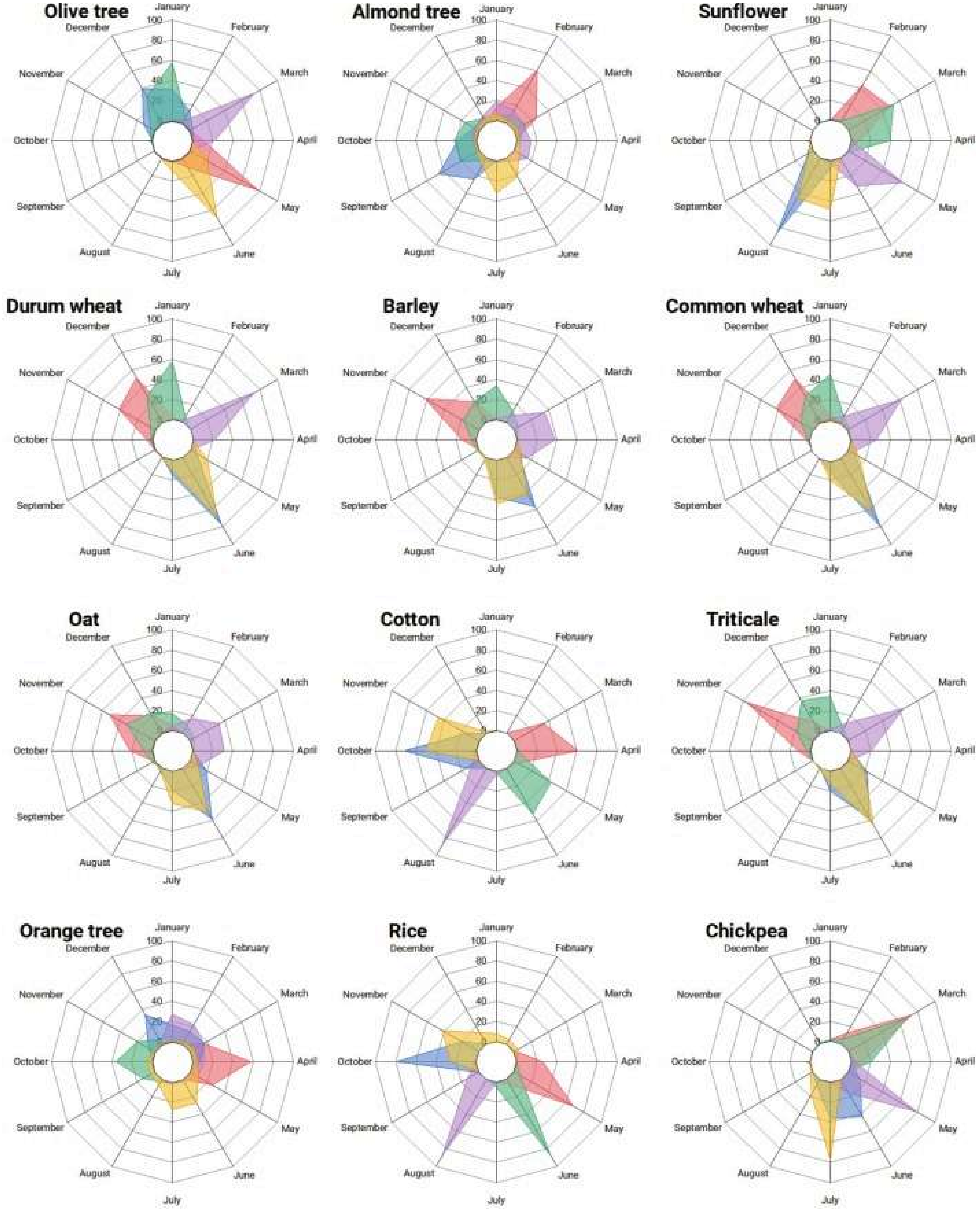

Olive tree	Almond tree	Sunflower	Durum wheat	Barley	Common wheat	Oat	Cotton	Triticale	Orange tree	Rice	Chickpea

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Fig. 8. Percentage of crop that was sown/flowered, harvested, and reached SOS, MOS and EOS in each month. In red, sowing/flowering; in blue, harvest; in green, SOS; in purple, MOS; in yellow, EOS. In the–– table, percentage of the crop that matched in the same month according to LSP and calendar data.

3.3. Crop grouping based on satellite-observed phenological patterns

The large diversity of crops in the study area prompted the use of a clustering algorithm, as one of the main objectives was to identify phenologically homogeneous agricultural zones. The results of the elbow method are shown in Fig. 9, where the analysis was conducted using between 1 and 10 clusters. For herbaceous crops, the total intra-cluster sum of distances decreased markedly up to three or four clusters (Fig. 9a), with a similar trend observed for woody crops (Fig. 9b). Beyond this point, increasing the number of clusters led to only marginal improvements in cohesion. When all crops were considered jointly, the analysis also suggested that three or four clusters would be optimal. However, due to the considerable heterogeneity in crop types, climatic conditions, altitude, soil characteristics, and agricultural practices across the region, six clusters were ultimately selected. This decision was supported by the interpretation of results pointing to three groups of herbaceous crops and three groups of woody crops. Although the MAPA crop calendar includes 12 crop families, this classification is primarily based on economic criteria—including, for example, categories such as "industrial crops"—and does not reflect phenological behavior. Therefore, the number of clusters was based on the results from the elbow method, which focused on phenological patterns.

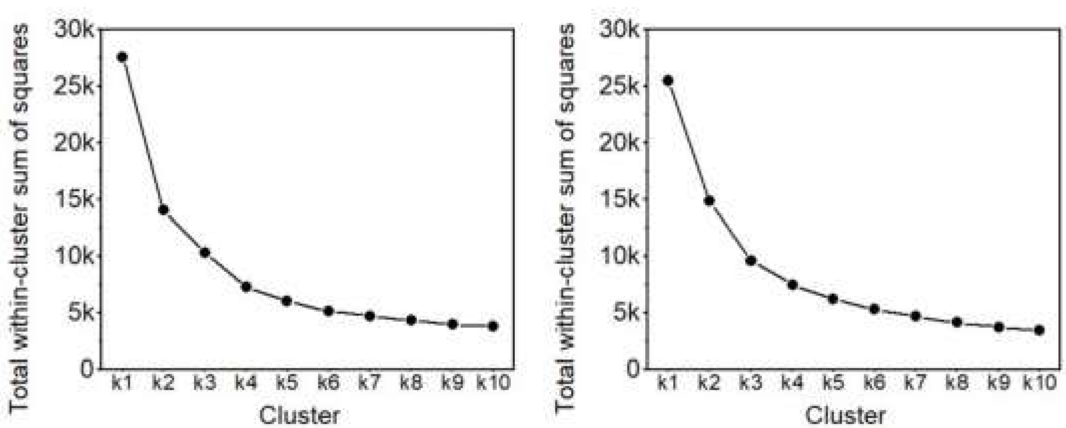

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Fig. 9. Total within-cluster sum of squares for clusters. From left to right, herbaceous crops and woody crops.

The clustering results, based on the sine and cosine transformations of SOS, MOS, and EOS (Table3), included all the crop categories listed in Table 1. Despite its relevance, the application of circular statistics in phenology remains limited in remote sensing studies. As noted by Fisher et al. (2007), phenological events exhibit strong interannual and spatial variability, reinforcing the importance of accounting for their cyclical nature. However, most large-scale analyses continue to rely on linear assumptions. Our use of sine and cosine transformations provides a robust alternative, as also proposed by Morellato et al. (2010), particularly in Mediterranean regions where the growing season does not always align neatly with the start or end of the calendar year. Most crops were predominantly assigned to a single main cluster. For instance, sunflower was grouped in cluster 6 in 92.1% of cases, and chickpea in 85% of cases. Cotton and rice were almost entirely assigned to cluster 2, with 97% and 99.8% respectively. A substantial proportion of cereal crops was grouped in cluster 1: 84.9% of durum wheat, 75% of common wheat, 67.2% of triticale, and 62.6% of barley. Nevertheless, a significant portion of these crops also appeared in cluster 4: 46.1% of oat, 30.6% of triticale, 27.1% of barley, 21.4% of common wheat, and 13.9% of durum wheat. Woody crops exhibited greater heterogeneity in their cluster assignments. Almond trees were mainly distributed between clusters 4 (37.6%) and 5 (28.5%), and a similar distribution was observed for orange trees, with 37.9% in cluster 4 and 33.7% in cluster 5. Olive trees were largely assigned to cluster 4 (64.3%), although a notable proportion (26.2%) was grouped in cluster 5. When the crop typology variable (herbaceous/woody) was included, most crop parcels continued to be concentrated within a single main cluster. Durum wheat and common wheat increased their assignment to a single cluster from 84.9% to 89.5%, and from 75% to 81%, respectively. Chickpea increased from 85% to 90.3%, barley from 62.6% to 68.9%, and triticale from 67.2% to 72.8%. Other crops, such as rice, sunflower, olive, oats, orange, and cotton, showed slight improvements in consistency. Almond was the only crop for which the concentration in its main cluster declined, although the decrease was minimal. Despite the complexity of the agricultural landscape, most clusters were largely associated with a single dominant crop type. Except for cluster 6, over 80% of the plots in each cluster belonged to a single category, with cluster 5 showing the highest internal homogeneity—nearly 100% of its area was associated with the same crop type. The inclusion of the herbaceous/woody variable provided limited benefits in the clustering process (Table 4). Therefore, the approach based solely on phenological variables was selected as shown in Fig. 10.

Table 3. Distribution of each crop by cluster (% of the area). Does not include herbaceous/woody

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Crop	k1 – Winter cereals and grain legumes	k2 – Spring-planted crops	k3 – Stone fruits and vineyards	k4- Olive groves, pastures, and mixed woody crops	k5- Subtropical and irrigated perennial crops	k6 – Summer vegetables and fruits
Olive tree	6.5	0.3	0.2	64.3	26.2	2.6
Almond tree	14.9	1.7	1.9	37.6	28.5	15.5
Sunflower	4.7	0.5	0.4	1.1	1.3	92.1
Durum wheat	84.9	0.1	0.1	13.9	0.1	1.0
Barley	62.6	0.2	0.1	27.1	1.3	8.8
Common wheat	75.0	0.1	0.0	21.4	0.4	3.0
Oat	43.7	0.8	0.2	46.1	1.7	7.5
Cotton	0.6	97.0	1.1	0.5	0.4	0.4
Triticale	67.2	0.2	0.1	30.6	0.7	1.3
Orange tree	11.3	1.6	2.2	37.9	33.7	13.4
Rice	0.1	99.8	0.1	0.0	0.0	0.0
Chickpea	10.4	0.0	0.2	2.2	2.3	85.0

Table 4. Distribution of each crop by cluster (% of the area). Includes herbaceous/woody variable.

Crop	k1 – Winter cereals and grain legumes	k2 – Spring-planted crops	k3 – Stone fruits and vineyards	k4- Olive groves, pastures, and mixed woody crops	k5- Subtropical and irrigated perennial crops	k6 – Summer vegetables and fruits
Olive tree	4.3	0.1	0.1	65.0	27.2	3.2

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Crop	k1 – Winter cereals and grain legumes	k2 – Spring-planted crops	k3 – Stone fruits and vineyards	k4- Olive groves, pastures, and mixed woody crops	k5- Subtropical and irrigated perennial crops	k6 – Summer vegetables and fruits
Almond tree	9.4	0.7	4.5	34.7	30.3	20.3
Sunflower	3.3	1.2	0.1	0.8	0.8	93.8
Durum wheat	89.5	0.1	0.1	9.8	0.0	0.6
Barley	68.9	0.2	0.0	21.1	1.0	8.9
Common wheat	81.0	0.1	0.0	15.6	0.7	2.6
Oat	47.0	0.5	0.1	43.9	1.4	7.2
Cotton	0.6	97.8	0.2	0.4	0.6	0.4
Triticale	72.9	0.2	0.1	24.8	0.6	1.5
Orange tree	8.5	0.8	4.4	41.4	28.5	16.5
Rice	0.0	99.9	0.0	0.0	0.0	0.0
Chickpea	7.1	0.0	0.1	1.2	1.3	90.3

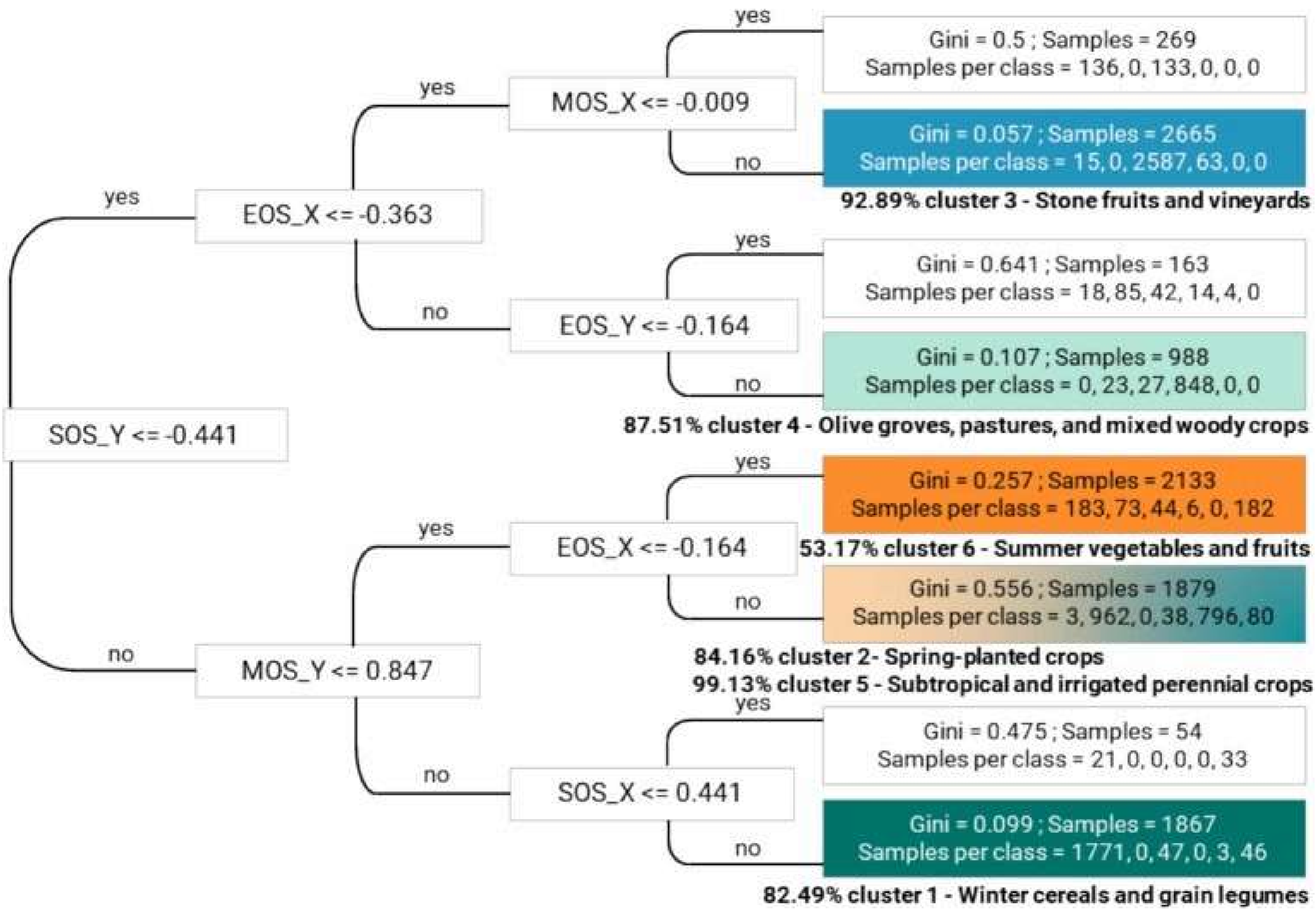

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Fig. 10. Km-CART model showing the different categories based on phenometrics. The gini coefficient of each final category is shown, as well as the number of samples per cluster in each final category and the largest percentage of every cluster included in each final category. The colours match cluster colours in figure 10 and they indicate the final category where most of the samples from each cluster were classified. Note that one category is the most representative for two different clusters. Cluster 1, Winter cereals and grain legumes; Cluster 2, Spring-planted crops; Cluster 3, Stone fruits and vineyards; Cluster 4, Olive groves, pastures, and mixed woody crops; Cluster 5, Subtropical and irrigated perennial crops; Cluster 6, Summer vegetables and fruits.

The temporal dynamics of the six phenological clusters identified in this study revealed clear patterns in crop seasonality and duration that align with both crop type (herbaceous vs. woody) and climatic conditions across Andalusia (Fig. 11). Rather than focusing solely on the individual

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practices. Clusters 1 and 2 encompass early-season crops with relatively short to medium growth cycles. Cluster 1, which accounts for 18.1% of the total study area, includes primarily winter cereals and grain legumes such as durum and common wheat, triticale, barley, oats, peas, and broad beans, sown in late autumn (around DOY 330) and harvested in spring or early summer (DOY 150–180). These crops dominate the Guadalquivir Valley and the Surco Intrabetico, where mild winters and moderate rainfall favour early development (Fig. 12). Cluster 2, occupying 3.2% of the area, represents other spring-planted crops with moderate cycles (DOY 10–130 to DOY 150–220), including a mix of irrigated vegetables like broccoli and lettuce, mostly found near major water bodies such as the Guadalquivir and Genil rivers. Clusters 3 and 4 group woody crops with longer cycles. Cluster 3, the smallest group with only 1.7% of the surface, contains mainly stone fruits and vineyards (e.g. peach, nectarine, plum, grapevine), typically budding in early spring (DOY 30–50) and harvested by early summer (DOY 180). Their limited extent reflects their specific climatic requirements and lower overall surface area. Cluster 4 is the most extensive, covering 54.4% of the study area. It comprises olive groves—by far the dominant land use in the region—along with pastures, fallow land, and mixed woody crops such as almond and fig trees. These crops tend to exhibit longer, more stable vegetative phases extending from late winter (DOY 50–80) through summer (DOY 200–250), particularly under rainfed conditions typical of Andalusia's Mediterranean interior. Clusters 5 and 6 are composed of later-season crops with growth cycles concentrated in the warmer months. Cluster 5, covering 11.5% of the surface, includes subtropical and irrigated perennial crops such as citrus (orange, lemon) and avocado, many of which initiate their phenological activity in early spring (DOY 90–120) and extend into late summer or early autumn (DOY 250–270). Cluster 6 accounts for 11.0% of the area and includes summer vegetables and fruits with short, intensive cycles—such as melon, watermelon, cherry, beetroot, and tomato for processing—planted in late spring (DOY 120–150) and harvested by late summer (DOY 240–270). These are heavily reliant on irrigation and are mainly distributed in areas with greater water access. From an applied perspective, the identification of these six phenological clusters offers valuable insights for agricultural management. For instance, clusters 1 and 2, which include early-season herbaceous crops with short to moderate cycles, may benefit from tailored input strategies (e.g., fertilization or weed control) adjusted to their rapid growth dynamics. In contrast, clusters 4 and 5, dominated by long-cycle woody crops, can guide irrigation planning or drought stress mitigation during critical summer months. These classifications can also inform regional agricultural policy, including the design of agro-environmental schemes or climate adaptation measures tailored to specific crop groups.

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Fig. 11. Clustering of crop clusters in concentric circles, from cluster 1, the innermost, to cluster 6, the outermost. Each bar within these clusters reflects the medians of key crop life cycle dates: the start of the bar marks the median SOS date, the change to a wider bar indicates the median MOS date, and the end of the bar represents the median EOS date. It does not include the herbaceous/dry variable. Cluster 1, Winter cereals and grain legumes; Cluster 2, Spring-planted crops; Cluster 3, Stone fruits and vineyards; Cluster 4, Olive groves, pastures, and mixed woody crops; Cluster 5, Subtropical and irrigated perennial crops; Cluster 6, Summer vegetables and fruits.

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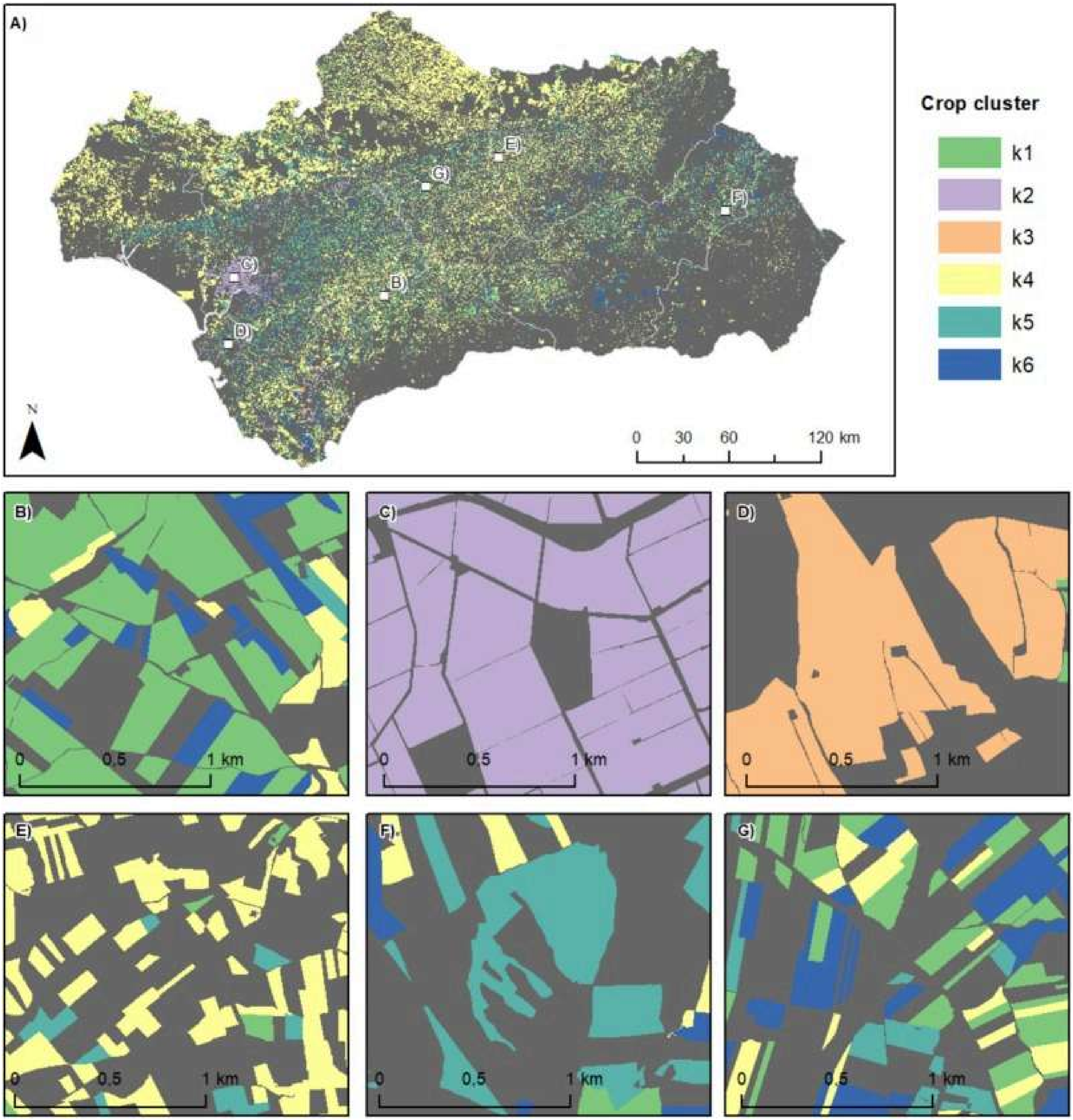

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Fig. 12. Distribution of crops in Andalusia by cluster. A) Andalusia; B) Cereal plots; C) Rice plots; D) Vineyards; E) Olive groves; F) Almond groves; G) Mixture of sunflower plots and other non-irrigated crops.

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The results obtained show that it is possible to produce crop calendars from Sentinel-2 remotely sensed data, as widely stated in literature (Vuolo et al., 2018, Maponya et al., 2020). The large amount of information available at a spatial scale of 10 m for such an extensive area shows the great potential of this methodology for the application of LSP in the elaboration of crop calendars. The availability of Sentinel-2 images at a global level and the replicable methodology of this study allows for an exhaustive characterisation of all crops with a significant presence in Andalusia. The availability of this information is highly valuable for researchers, public administrators, and economic markets, but it is particularly beneficial for farmers. This information can be very useful for making decisions regarding the management of their plots. The 10-metre spatial resolution allows for a detailed analysis including almost all plots, combining a large spatial extent with a medium–high spatial resolution.

The results diverge depending on the type of crop. Arable crops exhibit much stronger seasonality and show a high agreement between LSP data and field calendar records. The EOS-harvest ratio remains particularly high, while the percentage of each crop that is sown and reaches SOS in the same month is not that high, logically due to the delay between sowing and the time when the crop germinates, and SOS is distinguished through remote sensing. With respect to woody crops, the conclusions differ, since the variables collected by the field calendar for this type of crop, flowering and harvesting, do not have such a close relationship with SOS or EOS. Among the challenges presented by this methodology, it is worth highlighting the problem of the crops and categories included in each of the data sources. The MAPA crop calendar does not include those categories that are not strictly crops, leaving out covers with a wide presence in Andalusia, such as pastures in all their varieties or fallow land. On the other hand, the CAP does include this type of cover with agricultural, livestock and pastoral importance. However, there was the case of plots which, due to their economic nature, did not request subsidies from the CAP and were therefore excluded in the latter source, leaving a lack of information in some specific areas, especially in the strip of agriculture close to the coast, in which crops under plastic in Huelva in the west and Almeria in the east are of particular importance. The lack of data for these areas leaves an information gap in some key crops such as strawberries, fruits of the forest, flowers or vegetables.

The comparison of the LSP data with the data from an official product such as the MAPA crop calendars allows the results obtained to be validated. One of the main advantages of this source is its detailed results, expressed at the level of each of the NUTS-3 that make up the NUTS-2 of Andalusia. In addition, it is worth noting that, despite the existence of the Copernicus HR-VPP product, itis not specifically designed to capture crop phenophases, as it relies on model

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phenological metrics, enabling a more accurate characterization of agricultural cycles in regions such as Andalusia. The complementarity between both approaches highlights the potential for future integration, particularly as operational products continue to evolve toward greater sensitivity to agricultural systems. However, this source has several limitations. One limitation is the five-year update cycle of the results, which introduces a significant time lag between data collection and publication. In addition, the results express the percentage of the crop that is sown or harvested per month, so the temporal information is less than that obtained from LSP from Sentinel-2, every 5 days. In order to be able to compare both sources, the temporal detail of LSP is waived. Additionally, the database is aggregated at the NUTS-3 level, making validation at the plot level unfeasible. However, for the scale of this study, which aims to serve as a reference for phenology extraction in a large NUTS-2 region such as Andalusia, validation at the NUTS-3 level is considered to be adequate. These data-related constraints also have methodological implications, as they directly condition the type and robustness of the validation procedures that can be applied. One of the methodological aspects to be considered is the nature of the data, such as the DOY, which makes it difficult to establish a protocol. The problem lies in the fact that this data is usually treated as a linear variable, when it is a circular variable, which can make it difficult to process and make it operational. This issue becomes particularly relevant when attempting statistical comparisons across crops with markedly different phenological cycles, where standard linear validation approaches are not strictly appropriate. In this case it is of great importance, as it includes a multitude of crops with very varied phenological cycles, which do not necessarily develop completely within a calendar year. The proposed solution consists of decomposing each DOY into its sine and cosine, thus treating it as a circular variable.

5. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist in reviewing and improving the English language in the manuscript. After using this tool, the authors carefully reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The first author is a FPU grant holder funded by the Spanish "Ministerio de Universidades" (Reference FPU18/04366). The authors are grateful for the financial support given by the projects PID2023-152656OB-I00 and TSI-100930-2023-2, funded by "Ministerio de Ciencia, Innovación y Universidades de España" and "Ministerio de Asuntos Económicos y Transformación Digital", respectively.

Appendix A. Supplementary data

The following are the Supplementary data to this article:

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Supplementary Data 1.

Data availability

Data will be made available on request.

Common Agricultural Policy (CAP) data that support the findings of this study are available from the Spanish Agrarian Guarantee Fund (FEGA) upon request. Restrictions apply to the availability of these data, which were used under license for this study. As the data refer to individual farm information, we do not have permission to share them.

GISCAP data used in this study are openly available from FEGA at https://www.fega.gob.es/en ↗.

Official crop calendars used in this study are openly available from the Spanish Ministry of Agriculture, Fisheries and Food (MAPA) at:

https://www.mapa.gob.es/dam/mapa/contenido/estadisticas/temas/estadisticas-agrarias/2.agricultura/8.-calendarios_siembras-recoleccion/calendario-2014–2016/01-calendariosiembra-nuevo-sencilla-1.pdf.

Sentinel-2 satellite data used in this study are openly available from the Copernicus Open Access Hub.

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